

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech (ME) Examination

NOV 2018

ME371 Internal Combustion Engines

Date: 29.11.2018, Thursday

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the question below.

[05]

- a. The distance through which the piston can travel between TDC to BDC of cylinder is known as _____.
1. Bore 2. Stroke 3. Swept volume 4. Clearance volume
- b. Otto cycle is known as _____.
1. Constant volume air cycle 2. Constant Enthalpy air cycle
3. Constant Pressure air cycle 4. None of above
- c. An Ignition lag of 35 degree crank rotation of an engine running at 3000 rpm corresponds to _____.
1. 0.0002 sec 2. 0.002 sec 3. 0.00015 sec 4. 0.0015 sec
- d. Which of the following can cause engine Ping during acceleration?
1. Advanced ignition timing
2. Retarded ignition timing
3. Rich fuel mixture
4. None of above
- e. A vehicle that has a higher than normal HC and CO reading is most likely running _____.
1. Rich 2. Lean
3. At stoichiometric 4. Any of above

Q-2.(a) Discuss the general principles of S.I. engine combustion chamber design.

[05]

Q-2.(b) Taking variable specific heat into account, establish the relation [06]

$$\frac{de}{e} = \frac{dc_v}{c_v} \left[\frac{1-e}{e} (1-\gamma) \ln r \right]$$

For Otto cycle, where e is the efficiency and r is the compression ratio.

OR

(b) What do you understand by air-standard cycle? What are the assumptions made for analysis of air-standard cycle? [06]

(c) Discuss factors affecting Ignition lag in SI engines. [06]

OR

(c) Discuss Combustion stages of C. I. Engine in detail. [06]

Q-3.(a) What action can be taken with regard to following variable, in order to reduce the possibility of detonation of S.I. Engine? Justify your answers with reason. [07]

(i) Compression ratio (ii) Mixture inlet temperature (iii) Spark timing (iv) Turbulence

OR

(a) What are the effects of following variable on diesel knock? [07]

(i) Injection timing and rate of fuel injection (ii) Surface to volume ratio (iii) Turbulence

(b) Explain construction of rope brake dynamometer and derive expression for B.P. [06]

OR

(b) Explain with neat sketch working principle of NDIR. [06]

SECTION – II

Q - 4 Answer the question below. [05]

a. MPFI refers to

- | | |
|---------------------------------|--------------------------------|
| 1. Main Point Fuel Introduction | 2. Multi Point Fuel Injection |
| 3. Multi Point Fuel Induction | 4. Master Point Fuel Induction |

b. Firing order of a four cylinder engine is _____.

- | | | | |
|------------|------------|------------|------------|
| 1. 1-3-4-2 | 2. 1-3-2-4 | 3. 1-2-4-3 | 4. 2-3-1-4 |
|------------|------------|------------|------------|

c. With increase in self- ignition temperature of fuel in SI engine knocking tendency ____.

- | | | | |
|--------------|--------------|--------------------------------|------------------|
| 1. Decreases | 2. Increases | 3. Either increase or decrease | 4. None of above |
|--------------|--------------|--------------------------------|------------------|

- d.** During a cylinder leakage test (leak-down) air is noticed coming out of the oil fill opening. Which of the following is most likely the cause?
1. A defective head gasket
 2. A damaged or worn compression ring
 3. Intake valve is not sealing
 4. Exhaust valve is not sealing
- e.** Which of the following nozzles have a better cold starting performance
1. Single hole
 2. Multi hole
 3. Pintle
 4. Pintaux

Q-5.(a) Sketch and explain the fuel consumption loop in SI engine. **[07]**

OR

- | | | |
|------------|---|-------------|
| (a) | Discuss following systems of carburetors with neat sketches- (i) idling system (ii) choke (iii) economizer system | [07] |
| (b) | Explain working principle of battery ignition system with neat sketch. | [07] |

OR

- (b)** Describe different types of injection nozzles and discuss their relative advantages and disadvantages. **[07]**
- (c)** Describe the two types of general injection systems. Why the air injection system is not used nowadays? **[06]**

Q-6.(a) What are the various methods of turbocharging? Compare their relative merits and demerits. **[06]**

OR

- | | | |
|------------|--|-------------|
| (a) | Describe Miller supercharging system with neat sketches. | [06] |
| (b) | Explain working principle of pistonless engine | [04] |
